

ARSALAN QASIM

AI / ML Engineer | Final-Year BSAI Student

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PROFESSIONAL SUMMARY

Final-year Artificial Intelligence student at COMSATS University Islamabad with hands-on experience building end-to-end AI systems — from data collection pipelines and model training to deployment and automation. Proficient in PyTorch, HuggingFace Transformers, LangChain, and RAG pipelines, with practical exposure to Docker, FastAPI, and MLflow. Demonstrated ability to engineer production-level solutions with measurable impact, including a workflow automation tool that achieved a 90% reduction in processing time across 20+ international Shopify stores.

EDUCATION

Bachelor of Science in Artificial Intelligence (BSAI)

Expected January 2027

COMSATS University Islamabad | CGPA: 3.34 / 4.0

Relevant Certifications (DataCamp):

- Intermediate Deep Learning with PyTorch
- Supervised Learning with Scikit-learn

TECHNICAL SKILLS

Languages & Databases: Python, SQL, NoSQL

AI & Machine Learning: PyTorch, Scikit-learn, HuggingFace Transformers, LangChain, RAG Pipelines, NLP, Computer Vision, Deep Learning

MLOps & Deployment: Docker, FastAPI, MLflow, Streamlit

Data Engineering & Pipelines: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Selenium, BeautifulSoup, n8n, Make

APIs & Tools: OpenAI API, Anthropic API, Git & GitHub

PROJECTS

GlycoTwin: Neuro-Symbolic Digital Twin | Final Year Project (In Progress)

2024 – Present

- Developing a hybrid neuro-symbolic system for personalized glycemic response prediction using **LSTM/Transformer encoders** for metabolic pattern learning and **Neural ODEs with biological constraints** for physiologically realistic glucose dynamics modeling
- Implemented a Volumetric Nutrition Quantifier module using **FastSAM** for food segmentation and **Depth Anything V2** for depth estimation to enable portion-size and nutrition estimation from meal images
- Training on **OhioT1DM** and **Shanghai T2DM** clinical datasets alongside a self-curated Pakistani food density database
- Supervised by Dr. Basit Raza | Team of 2

Real-Time Driver Drowsiness Detection System

2024

github.com/arsalanqasim/real-time-driver-drowsiness-detection

- Built a hybrid computer vision pipeline fusing **fine-tuned YOLOv8s** (trained on 2,400 custom images with augmentation) and **MediaPipe Face Mesh** for real-time drowsiness detection via an OR-gate fusion architecture
- Achieved **95.2% Precision and 97.9% Recall** on a 500-image test set with **99.0 mAP50**, running at **~55 FPS on RTX 4060** — validated on live 30 FPS webcam with documented robustness against glasses reflections (94% accuracy)
- Implemented a 3-stage warning/SOS escalation protocol with a real-time **CustomTkinter** dashboard for live metrics and alarm controls

Neon DQN Drone Lab | Reinforcement Learning Course Project

2024

github.com/arsalangasim/neon-dqn-drone-lab

- Designed and built a **custom Pygame simulation environment** from scratch with a Gym-compatible interface, featuring dynamic urban obstacles and a cyberpunk 2D city layout
- Implemented a **Dueling/Double DQN** agent with a 3-layer CNN processing 6×32×32 observation tensors for autonomous drone navigation
- Diagnosed convergence failure caused by sparse reward signals; proposed curriculum learning solution requiring 1,500–3,000 episodes with gradually increasing environmental complexity

NLP GitHub Issue Triage Tool

2023

github.com/arsalangasim/triageiq

- Automated classification of **140,000+ raw repository labels** into a standardized taxonomy using NLP-based text classification, significantly reducing manual triaging effort

PROFESSIONAL EXPERIENCE

Automation Engineer | Upwork (E-commerce Client, Slovenia)

2024 – Present

- Built a **Streamlit-based automation application** that streamlined end-to-end Shopify product listing workflows across **20+ stores in 5+ European countries**
- Reduced per-product upload time from **45 minutes to 5 minutes (~90% reduction)** through iterative engineering — progressing from Selenium scripting to direct **Judge.me API integration**
- Implemented one-click **multilingual product localization** supporting automatic translation for country-specific stores across Italy, Germany, Greece, and others
- Delivered **3,000+ products** across multiple international stores over a 2+ year client engagement

ADDITIONAL INFORMATION

Competitive Programming: Active problem-solving practice on LeetCode, HackerRank, and Codewars

Languages: Urdu (Native) | English (B2)

Interests: Calisthenics